

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Name: EVENTS   | Comments:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Used: YES |
| EVENTS         | Out1 := D_LO Or D_LO_LO Or D_HI Or D_SP Or UNDERFLOW Or U_HI Or U_HI_HI Or U_LO Or U_SP Or OVERFLOW Or START Or STOP Or E_L1_D_LO Or E_L1_D_SAFE Or E_L1_U_HIHI Or E_L1_U_SAFE Or E_TS_D_SAFE Or E_TS_U_HIHI ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |
| End            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |
| Name: TEST     | Comments:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Used: NO  |
| TEST           | Out1 := START Or STOP ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |           |
| End            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |
| Name: NEW_S_AB | Comments:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Used: NO  |
| NEW_S_AB       | Temp1 := 0 ;<br>Temp2 := / Temp1 And ( D_LO Or D_LO_LO Or D_HI Or D_SP Or U_HI Or U_HI_HI Or U_LO Or U_SP Or OVERFLOW Or UNDERFLOW Or START Or STOP ) And SAB0 ;<br>Temp1 := Temp1 Or Temp2 ;<br>Temp3 := / Temp1 And SAB1 And ( BOMBA_LIGA Or BOMBA_DESLG ) ;<br>Temp1 := Temp1 Or Temp3 ;<br>SAB0 := ( SAB0 Or Temp3 Or SABINI ) And / Temp2 ;<br>SAB1 := ( SAB1 Or Temp2 ) And / Temp3 ;<br>Out1 := / SABINI ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |           |
| End            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |
| Name: G_BOMBA  | Comments:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Used: NO  |
| G_BOMBA        | Temp1 := 0 ;<br>Temp2 := BDESLIGADA And / D_BOMBA_LIGA And / Temp1 ;<br>Temp1 := Temp1 Or Temp2 ;<br>Temp3 := BLIGADA And / D_BOMBA_DESL And / Temp1 ;<br>Temp1 := Temp1 Or Temp3 ;<br>Temp4 := BDESLIGADA And / D_BOMBA_DESL And / Temp1 ;<br>Temp1 := Temp1 Or Temp4 ;<br>Temp5 := BLIGADA And / D_BOMBA_LIGA And / Temp1 ;<br>Temp1 := Temp1 Or Temp5 ;<br>Temp6 := ( BDESLIGADA Or Temp4 Or Temp3 Or B_INICIAL ) ;<br>Temp7 := BLIGADA Or Temp5 Or Temp2 ;<br>Temp8 := Temp2 Or Temp5 ;<br>Temp9 := Temp3 Or Temp4 ;<br>BDESLIGADA := Temp6 And / Temp8 ;<br>BLIGADA := Temp7 And / Temp9 ;<br>Out1 := / B_INICIAL ;<br>Out2 := Temp1 ;                                                                                                                                                                                                                                            |           |
| End            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |
| Name: SUP_AH   | Comments:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Used: NO  |
| SUP_AH         | Out1 := / SAH_INICIAL ;<br>Out2 := ST_SAH0 And ( E_H_DISABLE Or E_H_CTRL_T1 Or E_H_CTRL_T2 ) ;<br>Out3 := ST_SAH1 And ( START Or STOP Or OVERFLOW Or UNDERFLOW Or U_HI Or U_HI_HI Or U_LO Or U_SP Or D_LO Or D_LO_LO Or D_HI Or D_SP Or E_L1_U_SAFE Or E_L1_U_HIHI Or E_L1_D_SAFE Or E_L1_D_LO Or E_TS_U_HIHI Or E_TS_D_SAFE Or BOMBA_LIGA Or BOMBA_DESLG Or VREGULA Or VABRE Or VFECHA ) ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |           |
| End            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |
| Name: EL1H     | Comments:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Used: YES |
| EL1H           | Out1 := / SL1H_INICIAL ;<br>Out2 := ST_SL1H_LO And E_L1_U_SAFE ;<br>Out3 := ST_SL1H_SAFE And E_L1_U_HIHI ;<br>Out4 := ST_SL1H_SAFE And E_L1_D_LO ;<br>Out5 := ST_SL1H_HIHI And E_L1_D_SAFE ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |           |
| End            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |
| Name: EL2H     | Comments:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Used: YES |
| EL2H           | Out1 := / SL2H_INICIAL ;<br>Out2 := ST_SL2H_LOLO And U_LO ;<br>Out3 := ST_SL2H_SENS And D_LO_LO ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |           |
| End            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |
| Name: SPH      | Comments:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Used: YES |
| SPH            | Out1 := / SPH_INICIAL ;<br>Out2 := ST_SPH_POFF And BOMBA_LIGA ;<br>Out3 := ST_SPH_PON And BOMBA_DESLG ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |           |
| End            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |
| Name: SVH      | Comments:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Used: YES |
| SVH            | Out1 := / SVH_INICIAL ;<br>Out2 := ST_SVH_VC And ( VABRE Or VREGULA ) ;<br>Out3 := ST_SVH_VO And VFECHA ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |
| End            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |
| Name: SM1      | Comments:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Used: YES |
| SM1            | Out1 := ( ( SM_1 Or SM_3 Or SM_7 Or SM_28 Or SM_29 ) And STOP ) Or ( SM_4 And D_LO ) ;<br>Out2 := SM_0 And START ;<br>Out3 := ( ( SM_1 Or SM_3 ) And U_SP ) Or ( ( SM_16 Or SM_26 Or SM_27 ) And D_HI ) Or ( SM_5 And E_L1_U_HIHI ) ;<br>Out4 := ( SM_2 And D_LO ) Or ( SM_28 And U_LO ) Or ( SM_7 And U_HI_HI ) ;<br>Out5 := ( SM_16 Or SM_26 Or SM_27 Or SM_30 Or SM_2 Or SM_5 ) And STOP ;<br>Out6 := ( SM_2 And E_L1_D_SAFE ) Or ( SM_30 And D_HI ) ;<br>Out7 := ( SM_28 And ( E_TS_D_SAFE Or E_TS_U_HIHI ) ) Or ( SM_29 And E_L1_U_HIHI ) Or ( SM_3 And D_LO_LO ) ;<br>Out8 := ( SM_3 And E_L1_D_SAFE ) Or ( SM_5 And D_LO ) ;                                                                                                                                                                                                                                                    |           |
| End            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |
| Name: SM2      | Comments:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Used: YES |
| SM2            | Out1 := ( SM_14 And BOMBA_DESLG ) Or ( SM_15 And VFECHA ) Or ( ( SM_16 Or SM_26 Or SM_27 ) And ( E_TS_D_SAFE Or E_TS_U_HIHI ) ) Or ( SM_2 And U_HI_HI ) Or ( SM_30 And E_L1_U_HIHI ) ;<br>Out2 := ( SM_10 Or SM_6 ) And BOMBA_LIGA ;<br>Out3 := ( SM_6 Or SM_9 ) And BOMBA_DESLG ;<br>Out4 := ( SM_12 Or SM_6 ) And ( VFECHA Or VREGULA ) ;<br>Out5 := ( SM_11 Or SM_6 ) And VABRE ;<br>Out6 := SM_6 And ( E_H_CTRL_T1 Or E_H_CTRL_T2 Or E_H_DISABLE ) ;<br>Out7 := ( SM_25 And VFECHA ) Or ( SM_8 And BOMBA_LIGA ) ;<br>Out8 := ( SM_25 And BOMBA_DESLG ) Or ( SM_8 And ( VABRE Or VREGULA ) ) ;                                                                                                                                                                                                                                                                                      |           |
| End            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |
| Name: SM3      | Comments:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Used: YES |
| SM3            | Out1 := ( SM_26 And BOMBA_DESLG ) Or ( SM_27 And VFECHA ) Or ( SM_8 And ( E_H_DISABLE Or E_H_CTRL_T1 Or E_H_CTRL_T2 ) ) ;<br>Out2 := ( ( SM_11 Or SM_20 ) And BOMBA_LIGA ) Or ( ( SM_18 Or SM_9 ) And ( VFECHA Or VREGULA ) ) ;<br>Out3 := ( ( SM_12 Or SM_21 ) And BOMBA_LIGA ) Or ( ( SM_17 Or SM_9 ) And VABRE ) ;<br>Out4 := ( ( SM_13 Or SM_22 ) And BOMBA_LIGA ) Or ( SM_9 And ( E_H_DISABLE Or E_H_CTRL_T1 Or E_H_CTRL_T2 ) ) ;<br>Out5 := ( ( SM_10 Or SM_21 ) And ( VFECHA Or VREGULA ) ) Or ( ( SM_11 Or SM_17 ) And BOMBA_DESLG ) ;<br>Out6 := ( ( SM_10 Or SM_20 ) And VABRE ) Or ( ( SM_12 Or SM_18 ) And BOMBA_DESLG ) ;<br>Out7 := ( SM_10 And ( E_H_DISABLE Or E_H_CTRL_T1 Or E_H_CTRL_T2 ) ) Or ( ( SM_13 Or SM_19 ) And BOMBA_DESLG ) ;<br>Out8 := ( SM_11 And ( E_H_DISABLE Or E_H_CTRL_T1 Or E_H_CTRL_T2 ) ) Or ( ( SM_13 Or SM_24 ) And ( VREGULA Or VFECHA ) ) ; |           |
| End            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |

| Name:   | Comments:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Used: |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| SM4     | Out1 := ( SM_12 And ( E_H_DISABLE Or E_H_CTRL_T1 Or E_H_CTRL_T2 ) ) Or ( ( SM_13 Or SM_23 ) And VABRE ) ;<br>Out2 := ( SM_14 And ( VABRE Or VREGULA ) ) Or ( SM_15 And BOMBA_LIGA ) ;<br>Out3 := ( SM_14 And ( E_H_DISABLE Or E_H_CTRL_T1 Or E_H_CTRL_T2 ) ) Or ( SM_16 And BOMBA_LIGA ) ;<br>Out4 := ( SM_15 And ( E_H_DISABLE Or E_H_CTRL_T1 Or E_H_CTRL_T2 ) ) Or ( SM_16 And ( VABRE Or VREGULA ) ) ;<br>Out5 := ( SM_18 And ( E_H_DISABLE Or E_H_CTRL_T1 Or E_H_CTRL_T2 ) ) Or ( SM_19 And VABRE ) Or ( SM_24 And BOMBA_LIGA ) ;<br>Out6 := ( SM_7 And D_LO_LO ) ;<br>Out7 := ( ( SM_16 Or SM_26 Or SM_27 ) And E_L1_D_SAFE ) ;                                                                                                                                                                                                                                        | YES   |
| End     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |
| COORD   | Out1 := ( C_1 Or C_2 Or C_3 Or C_4 ) And STOP ;<br>Out2 := ( C_0 And U_SP ) Or ( ( C_2 Or C_4 ) And D_HI ) Or ( C_3 And U_LO ) ;<br>Out3 := ( C_1 And U_HI_HI ) Or ( C_4 And E_L1_U_HIHI ) ;<br>Out4 := C_1 And D_LO_LO ;<br>Out5 := C_2 And E_L1_D_SAFE ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | YES   |
| End     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |
| SAB     | Out1 := SAB0 And ( BOMBA_LIGA Or BOMBA_DESLG ) ;<br>Out2 := SAB1 And ( START Or STOP Or UNDERFLOW Or D_LO_LO Or D_LO Or D_HI Or D_SP Or U_LO Or U_SP Or U_HI Or U_HI_HI Or OVERFLOW Or E_L1_U_SAFE Or E_L1_U_HIHI Or E_L1_D_SAFE Or E_L1_D_LO Or E_TS_U_HIHI Or E_TS_D_SAFE ) ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | NO    |
| End     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |
| SAV     | Out1 := SAV0 And ( VFECHA Or VABRE Or VREGULA ) ;<br>Out2 := SAV1 And ( D_LO Or D_LO_LO Or D_HI Or D_SP Or UNDERFLOW Or U_HI Or U_HI_HI Or U_LO Or U_SP Or OVERFLOW Or START Or STOP Or E_L1_U_SAFE Or E_L1_U_HIHI Or E_L1_D_SAFE Or E_L1_D_LO Or E_TS_U_HIHI Or E_TS_D_SAFE ) ;                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | NO    |
| End     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |
| DISABIL | Out1 := SAV1 Or SIBV_0 Or SIBV_1 Or ST_SL1H_LO Or ST_SL1H_SAFE Or C_3 Or SM_0 Or SM_1 Or SM_5 Or SM_6 Or SM_11 Or SM_13 Or SM_15 Or SM_21 Or SM_24 Or SM_27 Or SM_30 ;<br>Out2 := SAV1 Or SIBV_2 Or C_3 Or SM_1 Or SM_6 Or SM_11 Or SM_15 Or SM_21 Or SM_24 Or SM_27 Or SM_30 ;<br>Out3 := SAV1 Or SM_0 Or SM_5 Or SM_13 Or SM_27 ;<br>Out4 := SAB1 Or SIBV_0 Or C_2 Or SM_0 Or SM_5 Or SM_14 Or SM_28 ;<br>Out5 := SAB1 Or C_3 Or SM_1 Or SM_6 Or SM_11 Or SM_15 Or SM_21 Or SM_28 Or SM_29 Or SM_30 ;<br>Out6 := ST_SAH0 Or ST_SL1H_LO Or ST_STS_HIHI Or SM_0 Or SM_5 Or SM_12 Or SM_22 Or SM_26 Or SM_25 ;<br>Out7 := ST_SVH_VC Or ST_SAH1 Or ST_SL1H_LO Or ST_SL1H_SAFE Or ST_SL2H_LOLO Or ST_SPH_POFF Or ST_STS_HIHI Or SM_0 Or SM_5 Or SM_6 Or SM_11 Or SM_12 Or SM_15 Or SM_21 Or SM_22 Or SM_25 Or SM_26 ;<br>Out8 := ST_SAH1 Or SM_12 Or SM_22 Or SM_25 Or SM_26 ; | YES   |
| End     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |

































































